# MGT 3213 - Management Information Systems

## S # 8: Data Communications and Networking


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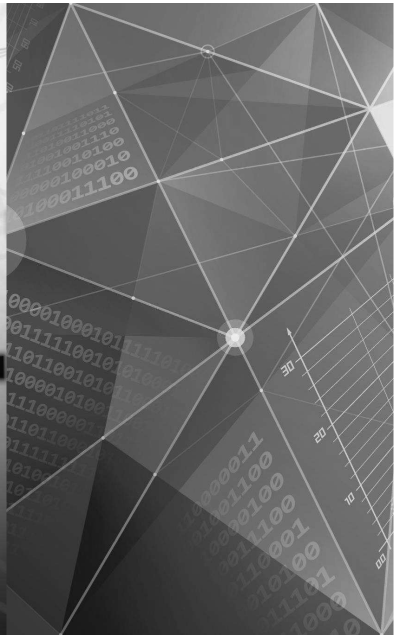
## Content

- Management Information Systems
- Information Systems (IS) and Uses
- Gaining Competitive Advantage with Information Systems
- The Strategic Role of Information Systems
- Information Systems Planning
- Information Systems Development
- The Database Approach to Data Management
- Data Communications and Networking

# Data Communications and Networking



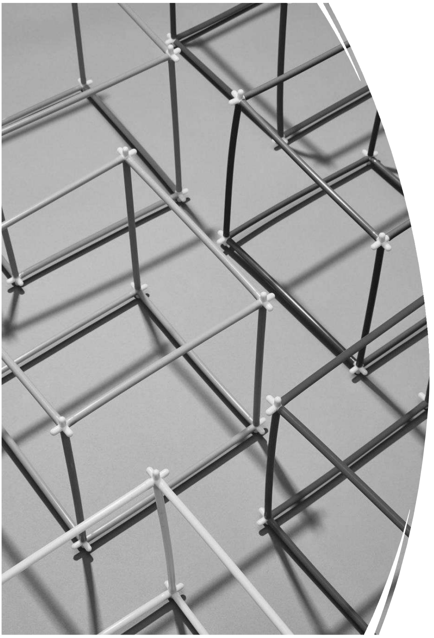
## Data Communications and Networking

- Introduction to Data Communications
- Advantages and Disadvantages of the Data Communications and Networking
- Network Models (Data Communications and Networking)
- Network Standards (Data Communications and Networking)
- Future Trends (Data Communications and Networking)

# Introduction To Data Communications

- Data Communication is a **process of exchanging data** or information.
- In case of computer networks this exchange is done between two devices over a transmission medium.
- This process involves a communication system which is made up of hardware and software.

What are the advantage of the Data Communications and Networking?



# Advantage of the Data Communications and Networking

- **Data Sharing:**
  - Networking enables seamless sharing of data and information among connected devices and systems, fostering collaboration and improving overall productivity.
- **Resource Sharing:**
  - Networks allow the sharing of hardware resources such as printers, scanners, and storage devices.
  - This reduces costs and promotes efficient use of resources across an organization.
- **Remote Access and Mobility:**
  - With networking, users can access data and applications remotely, facilitating flexible work arrangements and improving mobility.
  - This is especially crucial in today's globalized and mobile workforce.

# Advantage of the Data Communications and Networking

- **Cost Efficiency:**
  - Shared resources, centralized management, and the ability to leverage economies of scale contribute to cost efficiency in data communications and networking compared to individualized solutions.
- **Communication Speed:**
  - Networking technologies provide fast and efficient communication, enabling rapid data transfer and real-time collaboration.
  - This is essential for applications such as video conferencing and online collaboration tools.
- **Scalability:**
  - Networks can be easily scaled to accommodate a growing number of users, devices, and applications.
  - This flexibility allows organizations to adapt to changing business requirements without significant infrastructure changes.

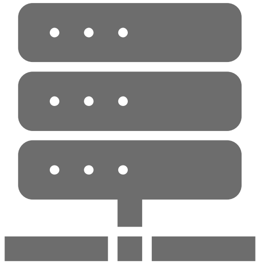
## Advantage of the Data Communications and Networking

**Centralized Management:**

- Networking allows for centralized management of resources, security policies, and software updates.
- This streamlines administration tasks and ensures consistency across the network.

**Increased Accessibility:**

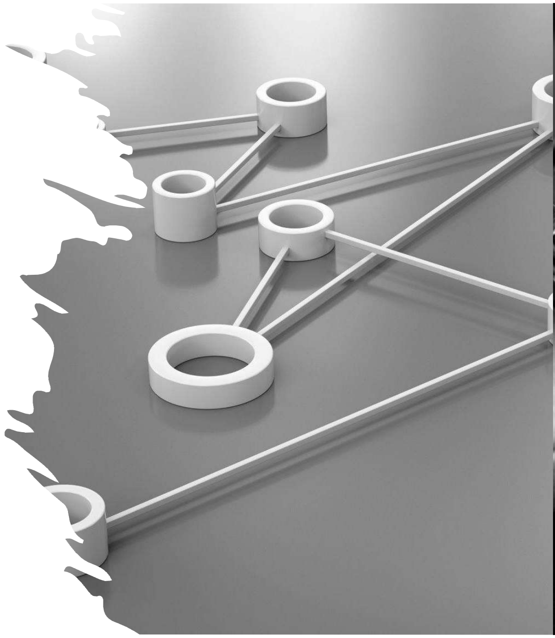
- Networks make information more accessible to authorized users, regardless of their physical location.
- This accessibility enhances information retrieval and decision-making processes.



## Advantage of the Data Communications and Networking

- **Enhanced Reliability and Redundancy:**
  - Networking technologies often incorporate redundancy and failover mechanisms, reducing the risk of downtime.
  - This ensures continuous access to critical resources and applications.
- **Global Connectivity:**
  - Networks enable global connectivity, facilitating communication and data exchange on a global scale.
  - This is particularly important for businesses operating in multiple locations or collaborating with partners worldwide.

## What are the Disadvantage of the Data Communications and Networking?



## Disadvantage of the Data Communications and Networking

- **Security Concerns:**
  - With the increased transfer of sensitive information over networks, the risk of unauthorized access, data breaches, and cyber-attacks becomes a significant concern.
- **Cost:**
  - Implementing and maintaining a robust network infrastructure can be expensive.
  - This includes the cost of hardware, software, security measures, and skilled personnel.
- **Complexity:**
  - Networking technologies can be complex, requiring specialized knowledge and expertise.
  - Managing and troubleshooting network issues may pose challenges for organizations without skilled IT personnel.

# Disadvantage of the Data Communications and Networking

- **Dependence on Infrastructure:**
  - Reliable data communication relies heavily on infrastructure components such as cables, routers, and servers.
  - Any failure in these components can disrupt communication and business operations.
- **Bandwidth Limitations:**
  - Network congestion and limited bandwidth can lead to slower data transfer speeds and degraded performance, especially in crowded or high-traffic environments.
- **Compatibility Issues:**
  - Ensuring compatibility between different devices, protocols, and software can be challenging.
  - Incompatibility issues may arise when integrating new technologies or when dealing with diverse systems.

# Disadvantage of the Data Communications and Networking

**Vulnerability to Viruses and Malware:**

- Networks are susceptible to the spread of viruses and malware. A single infected device can potentially compromise the entire network if not properly secured.

**Privacy Concerns:**

- As data is transmitted over networks, there is a risk of interception and unauthorized access, raising concerns about user privacy and confidentiality.

# Disadvantage of the Data Communications and Networking

**Maintenance and Upkeep:**

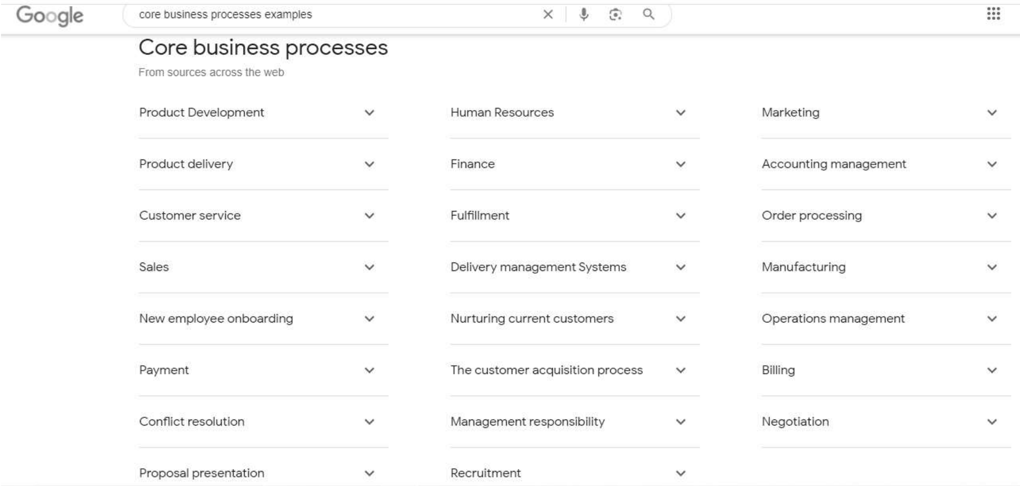
- Regular maintenance is crucial to ensure the smooth functioning of network components. Failure to perform routine updates and maintenance tasks can lead to system vulnerabilities and decreased performance.

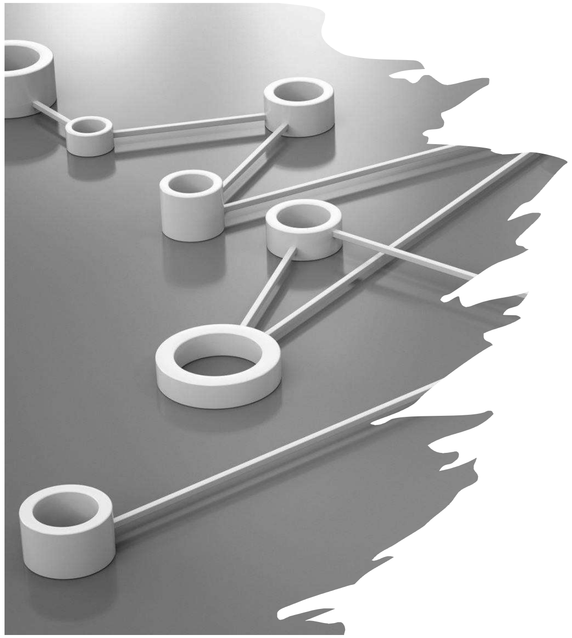
**Downtime and Service Disruptions:**

- Network outages, whether due to technical issues, maintenance, or external factors, can result in significant downtime. This can impact business operations and customer satisfaction.

# How are computer networks utilized in MIS?

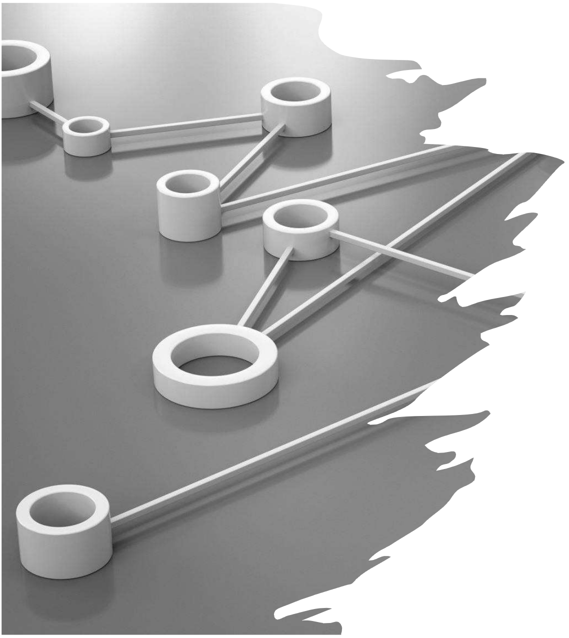
## How are computer networks utilized in MIS?



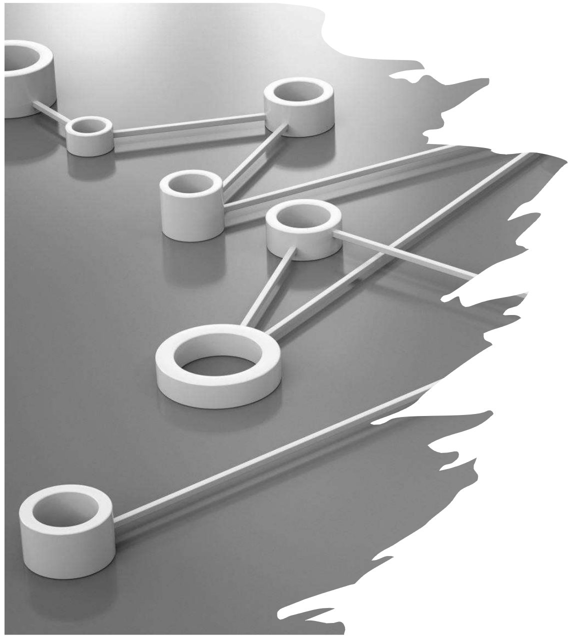
## The use of computer networks in Management Information Systems

- **Integration of Business Processes:**
  - MIS relies on computer networks to integrate various business processes seamlessly.
  - This integration enhances efficiency, reduces redundancy, and improves the overall performance of the organization.



## The use of computer networks in Management Information Systems

- The use of computer networks in Management Information Systems (MIS) is crucial for:
  - enhancing communication,
  - collaboration, and
  - information flow within an organization.

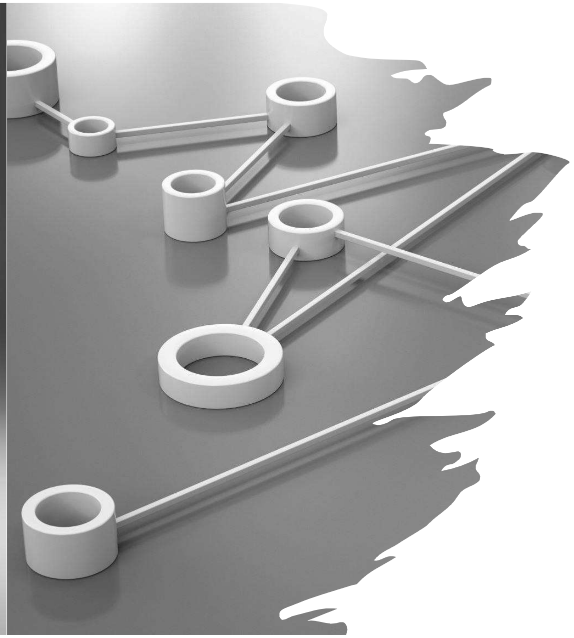


## How are computer networks utilized in MIS?

- **Data Sharing and Collaboration:**
  - Sharing of data and information among various departments and employees.
  - This facilitates collaboration and ensures that everyone has access to the necessary information for decision-making.
- **Centralized Data Storage**:
  - Computer networks allow for centralized storage of data.
  - This ensures that information is stored securely in a central location, making it easier to manage, back up, and retrieve when needed

# How are computer networks utilized in MIS?

- **Remote Access:**
  - Networks enable remote access to MIS, allowing authorized personnel to access information and applications from different locations.
  - This is particularly beneficial for organizations with distributed teams or employees working remotely.
- **Real-time Communication:**
  - Computer networks support real-time communication through emails, instant messaging, video conferencing, and other collaborative tools.
  - This ensures that information is communicated promptly, facilitating quicker decision-making processes.

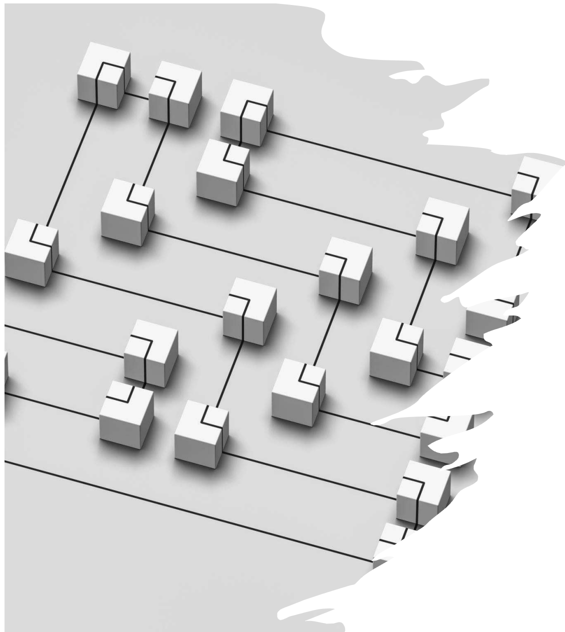


# How are computer networks utilized in MIS?

- **Resource Sharing:**
  - Networked MIS allows for the sharing of hardware resources, such as printers, scanners, and storage devices.
  - This helps in optimizing resource utilization and reducing costs.
- **Automated Data Processing:**
  - Computer networks enable the automation of data processing tasks within the MIS.
  - This includes data entry, validation, processing, and reporting, leading to increased efficiency and reduced errors.

# How are computer networks utilized in MIS?

- **Information Security:**
  - Networks play a crucial role in ensuring the security of information within the MIS.
  - Access controls, encryption, and other security measures are implemented to protect sensitive data from unauthorized access.
- **Decision Support Systems (DSS):**
  - Computer networks support the implementation of Decision Support Systems, which assist management in making informed decisions.
  - These systems analyze data from various sources and provide insights that aid in decision-making.

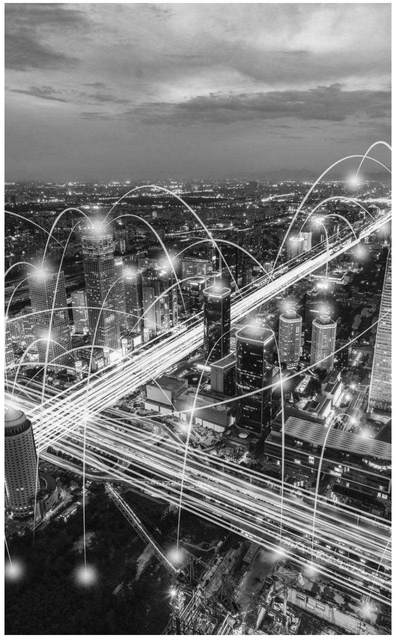


# How are computer networks utilized in MIS?

- **Enterprise Resource Planning (ERP):**
  - ERP systems, a type of MIS, leverage computer networks to integrate and manage core business processes, such as finance, human resources, supply chain, and more, across an entire organization.

## How are computer networks utilized in MIS?

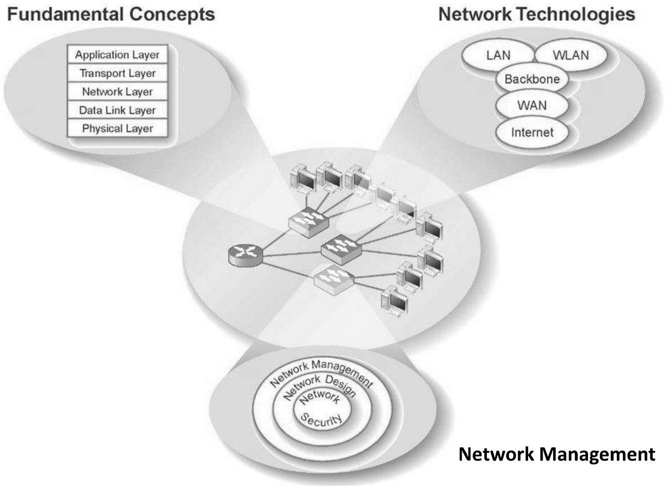
- **Customer Relationship Management (CRM):**
  - CRM systems within MIS use computer networks to collect, manage, and analyze customer data.
  - This helps in building and maintaining strong relationships with customers.
- **Monitoring and Reporting:**
  - Computer networks support the monitoring of activities within the MIS.
  - Managers can generate reports and analyze key performance indicators to track the organization's performance and identify areas for improvement.



## Computer networks in MIS

- Computer networks are foundational to the effective functioning of Management Information Systems,
  - providing the infrastructure for seamless communication,
  - data sharing, and
  - automation of processes within an organization.

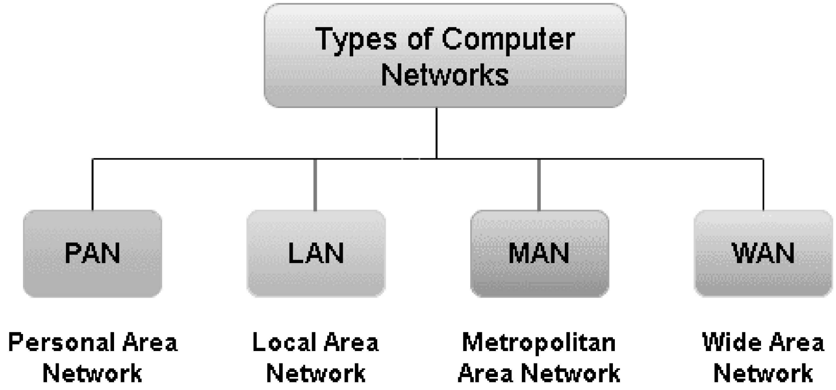
## The Three Faces of Networking

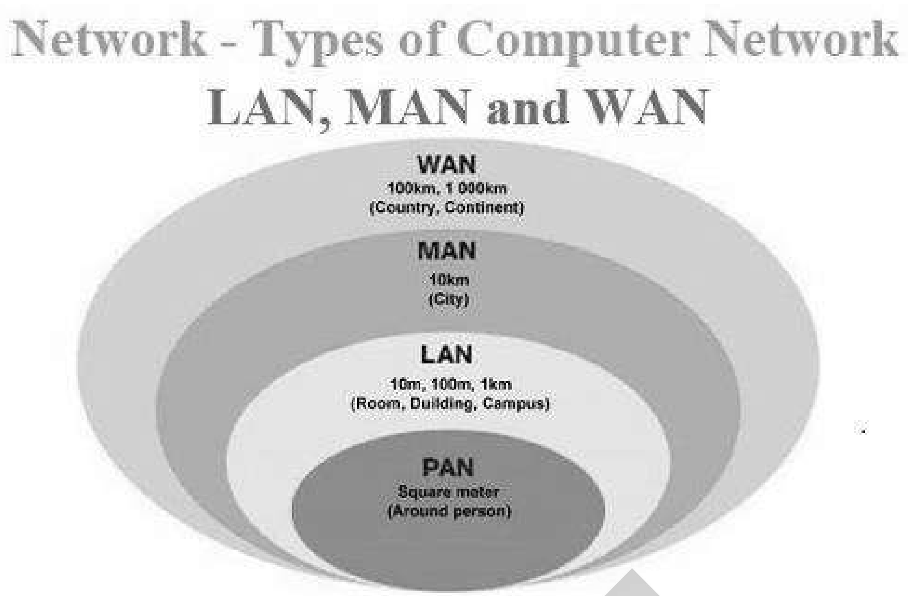



## Network Technologies

# Computer Network Types

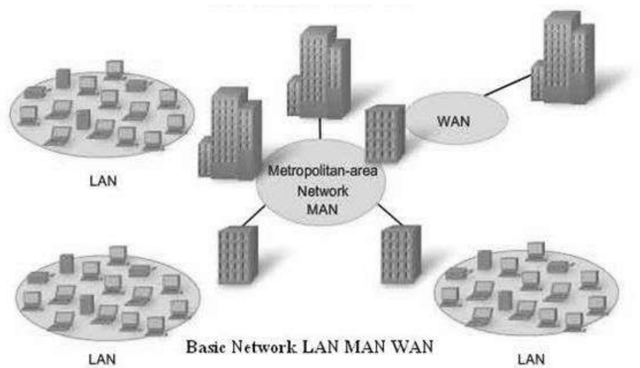
- A computer network is a group of computers linked to each other that enables the computer to communicate with another computer and share their resources, data, and applications.
- A computer network can be categorized by their size.
- A **computer network** is mainly of **four types**:
  - PAN (Personal Area Network)
  - LAN (Local Area Network)
  - MAN (Metropolitan Area Network)
  - WAN (Wide Area Network)




## Network Technologies - Network Topologies

- Local Area Network (LAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)

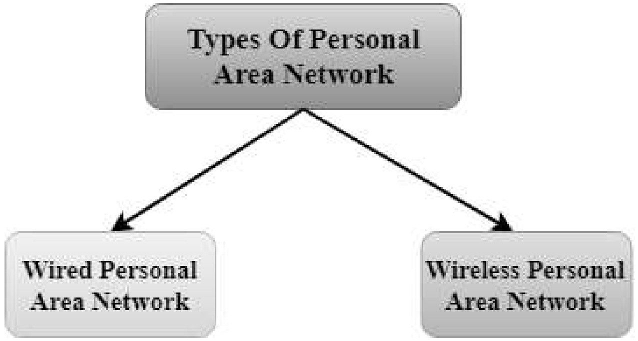


# PAN(Personal Area Network)

- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters (30 feet).

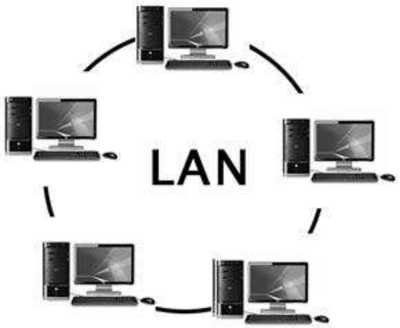
## There are two types of Personal Area Network:

- Wired Personal Area Network
  - Wireless Personal Area Network is developed by simply using wireless technologies such as WiFi, Bluetooth
- Wireless Personal Area Network
  - Wired Personal Area Network is created by using the USB

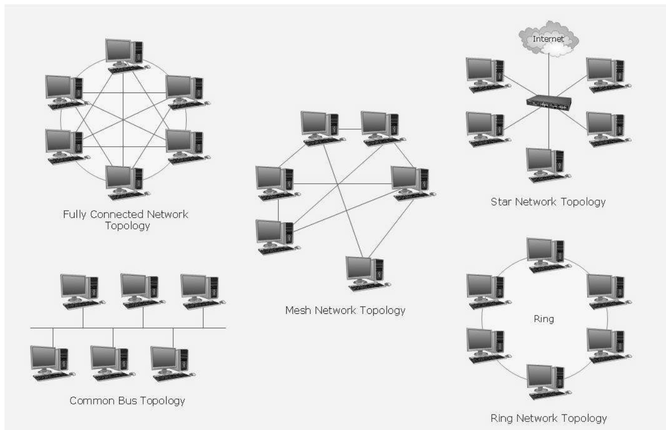



# LAN(Local Area Network)

- LANs are networks usually confined to a geographic area, such as a single building or a college campus.
- LANs can be small, linking as few as three computers, but often link hundreds of computers.

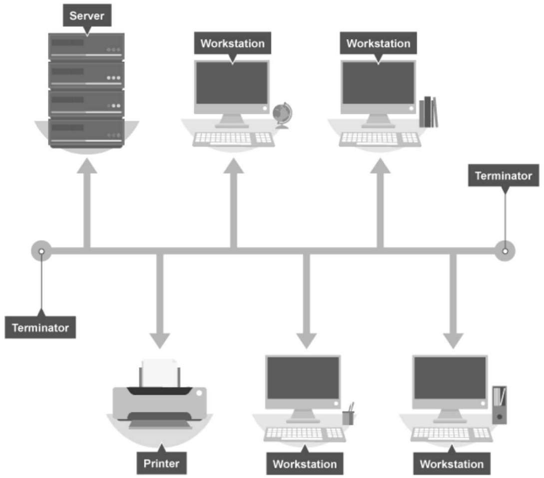



# LAN Topology



# The Bus Network

- In a bus network all the **workstations**, **servers** anetwork printers are joined to one cable (the bus).
- At each end of the cable a **terminator** is fitted to stop signals reflecting back down the bus.




# Advantages and disadvantages of a Bus Network

**The advantages of a bus network are:**

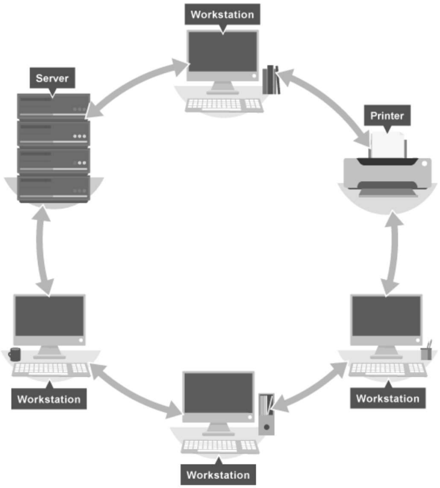
- It is easy to install
- It is cheap to install, as it doesn't require much cable

**The disadvantages of a bus network are:**

- If the main cable fails or gets damaged the whole network will fail
- As more workstations are connected the performance of the network will become slower because of data collisions
- Every workstation on the network "sees" all the data on the network – this is a security risk

# The Ring Network

- In a ring network each device (**workstation**, **server**, printer) is connected to two other devices - this forms a ring for the signals to travel around.
- Each packet of **data** on the network travels in one direction and each device receives each packet in turn until the destination device receives it.



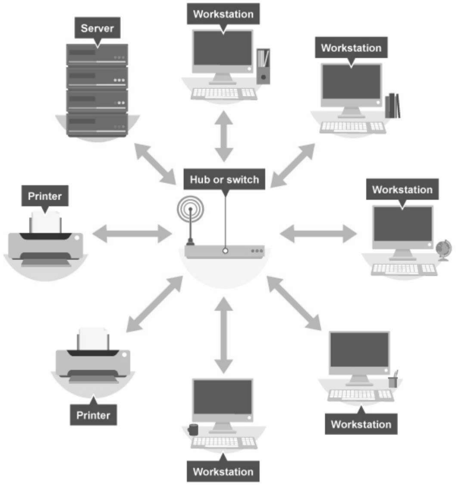

# Advantages and disadvantages of a Ring Network

This type of network can transfer data quickly, even if there are many devices connected because the data only flows in one direction, so there won't be any data collisions.

However, the real disadvantage is that if the main cable fails or any device is faulty, then the whole network will fail.

# The Star Network

- In a star network each device on the network has its own cable that connects to a **switch** or hub.
- A hub sends every packet of **data** to every device, whereas a switch only sends a packet of data to the destination device.




# Advantages and disadvantages of a Star Network

The advantages of a star network are:

- it is very reliable – if one cable or device fails then all the others will continue to work
- it is high-performing as no data collisions can occur

The disadvantages of a star network are:

- it is expensive to install as this type of network uses the most cable (network cable is expensive)
- extra hardware is required (hubs or switches) which adds to cost
- if a hub or switch fails, all the devices connected to it will have no network connection

# Network Topologies

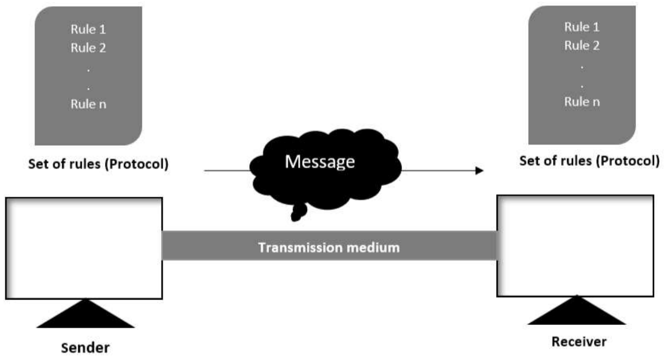
- Metropolitan Area Network (MAN):
  - MAN is basically a bigger version of a LAN and normally uses similar technology.
  - It might cover a group of near by corporate offices or a city and might be either private or public.

# Network Topologies

- Wide Area Network (WAN):
  - Often a network is in multiple physical places.
  - Wide area networking combines multiple LANs that are geographically separate.
  - This is accomplished by connecting the different LANs using services such as dedicated leased phone lines, dial-up phone lines (both synchronous and asynchronous), satellite links, and data packet carrier services.

## Data Communication System Components

- There are mainly five components of a data communication system:
  1. Message
  2. Sender
  3. Receiver
  4. Transmission Medium
  5. Set of rules (Protocol)



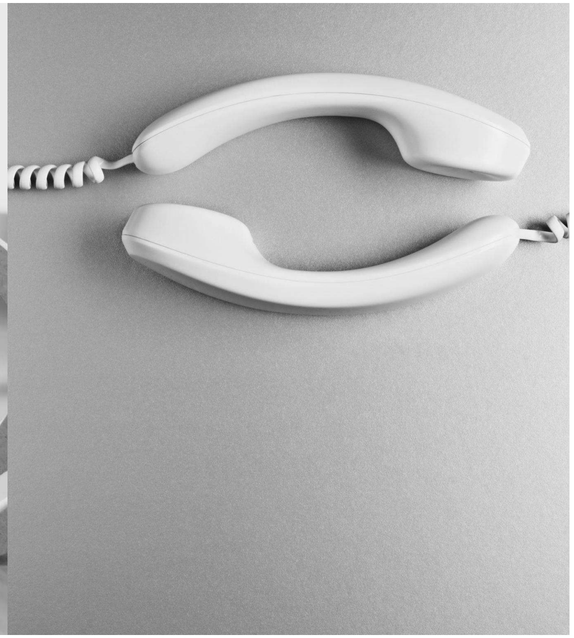

# Data Communication System Components

- **Message**
  - This is most useful asset of a data communication system.
  - The message simply refers to data or piece of information which is to be communicated.
  - A message could be in any form, it may be in form of a text file, an audio file, a video file, etc

# Data Communication System Components

### Sender

- To transfer message from source to destination, someone must be there who will play role of a source.
- Sender plays part of a source in data communication system.
- It is simple a device that sends data message.
- The device could be in form of a computer, mobile, telephone, laptop, video camera, or a workstation.



## Data Communication System Components

### Receiver

- It is destination where finally message sent by source has arrived.
- It is a device that receives message.
- Same as sender, receiver can also be in form of a computer, telephone mobile, workstation, etc.

## Data Communication System Components

### Transmission Medium

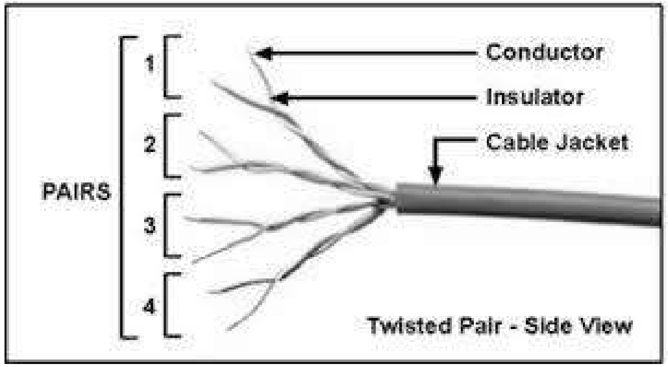
- Transmission Medium act as a bridge between sender and receiver.
- Transmission medium could be guided (with wires) or unguided (without wires), for example, twisted pair cable, fiber optic cable, radio waves, microwaves, etc.

## Data Communication System Components

**Types of Transmission Media**

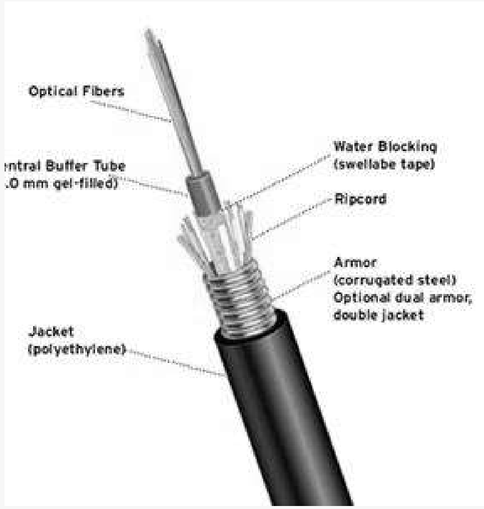
- Guided Media
  - Twisted Pair Cable
  - Coaxial Cable
  - Optical Fibre Cable
- Unguided Media
  - Radiowaves
  - Microwaves
  - Infrared

## Twisted pair wire





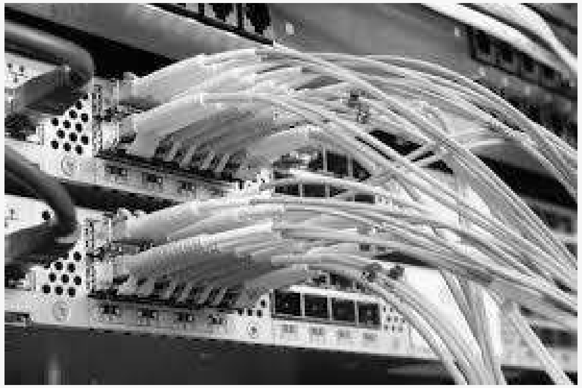

## Coaxial cable




## Opticl Fibre cable

- It uses the concept of reflection of light through a core made up of glass or plastic.
- The core is surrounded by a less dense glass or plastic covering called the cladding.
- It is used for transmission of large volumes of data.



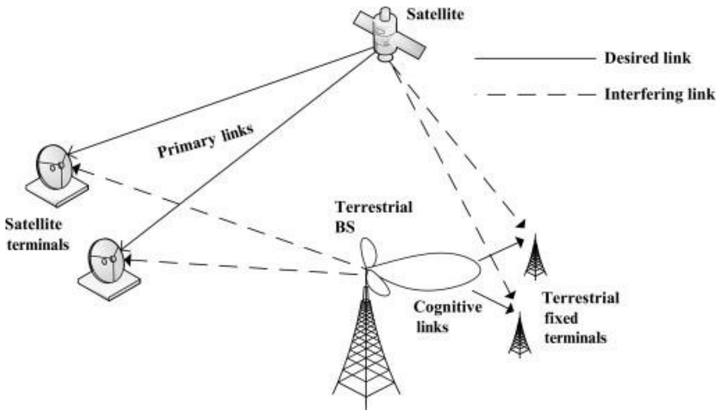


## Opticl Fibre cable

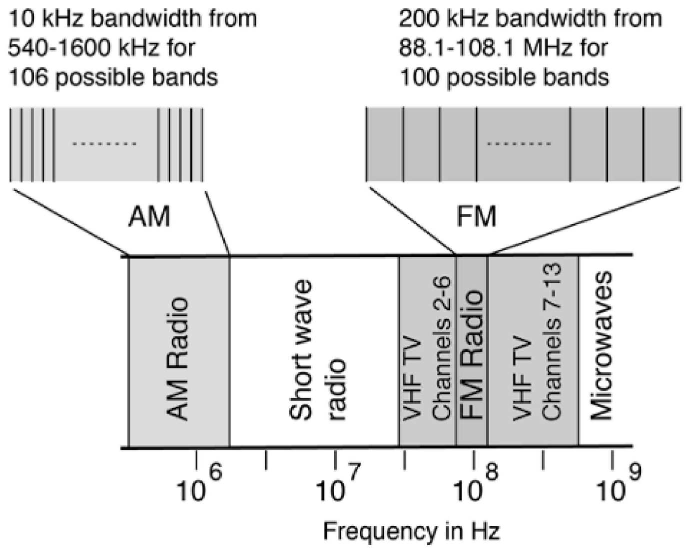
- **Advantages**:
  - Increased capacity and bandwidth
  - Light weight
  - Less signal attenuation
  - Immunity to electromagnetic interference
  - Resistance to corrosive materials
- **Disadvantages:**
  - Difficult to install and maintain
  - High cost
  - Fragile

## Unguided Media

- **Radiowaves –** These are easy to generate and can penetrate through buildings.
- The sending and receiving antennas need not be aligned.
- Frequency Range:
  - 3KHz – 1GHz.
  - AM and FM radios and cordless phones use Radiowaves for transmission.
- Further Categorized as
  - (i) Terrestrial and
  - (ii) Satellite.




# AM and FM Radio Frequencies

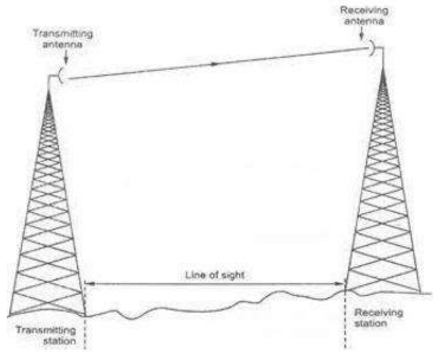


## Unguided Media

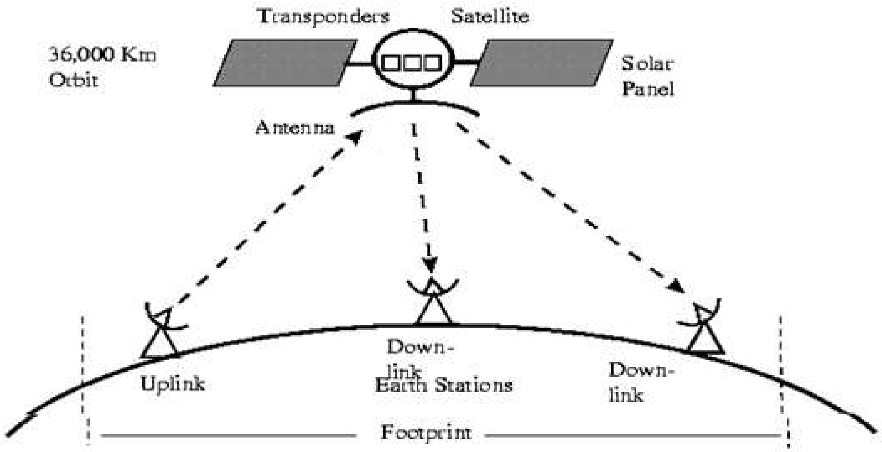
- **Microwaves**
  - It is a line-of-sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other.
  - The distance covered by the signal is directly proportional to the height of the antenna. Frequency Range:1GHz – 300GHz.
  - These are majorly used for mobile phone communication and television distribution.

## **Applications of Micro Waves**

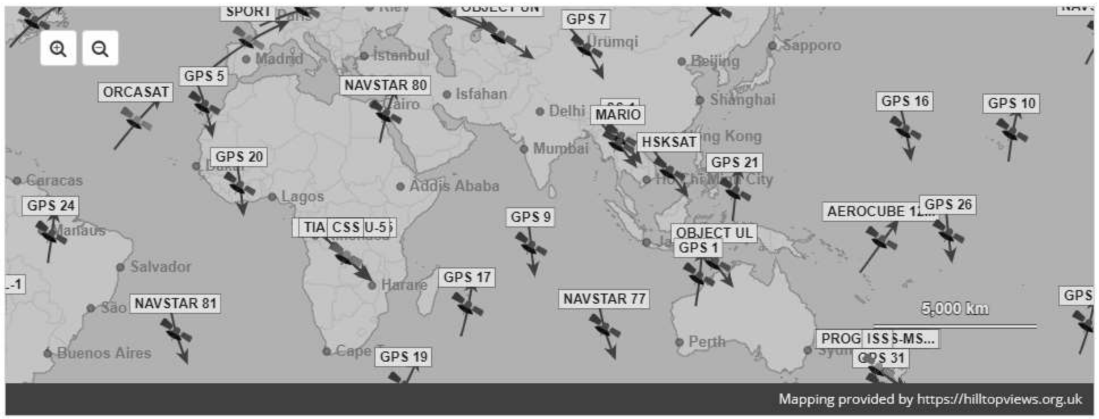
- Due to the unidirectional properties of Micro Waves, they are very useful when unicast.
- One-to-one communication is needed between the sender and the receiver.
- Cellular phones, satellite networks, and wireless LANs are using Micro Waves.




## Unguided Media - **Satellite**




## Live Map of Satellite Positions




Taken from - https://in-the-sky.org/satmap_worldmap.php

# Unguided Media

- **Infrared**
  - Infrared waves are used for very short distance communication.
  - They cannot penetrate through obstacles.
  - This prevents interference between systems.
  - Frequency Range:300GHz – 400THz.
  - It is used in TV remotes, wireless mouse, keyboard, printer, etc.

# Data Communication System Components

**Set of rules (Protocol)** :

- The protocol is a set of rules that govern data communication.
- If two different devices are connected but there is no protocol among them, there would not be any kind of communication between those two devices.
- Thus, the protocol is necessary for data communication to take place.

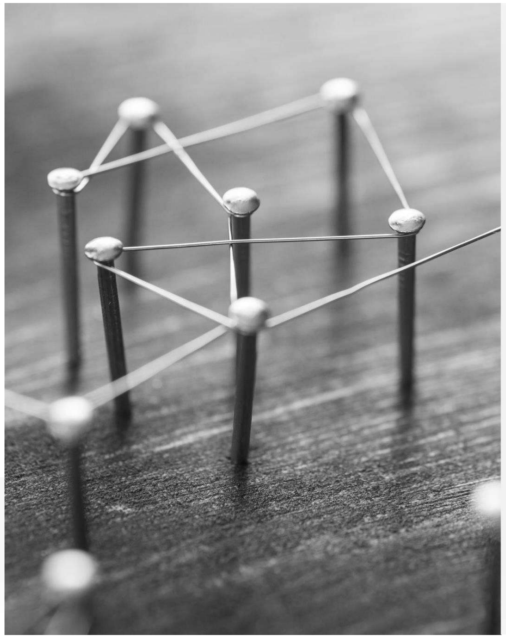
Examples of Protocal

- TCP/IP (Transmission Control Protocol/Internet Protocol)
- ARP (Address Resolution Protocol)
- DHCP (Dynamic Host Configuration Protocol)
- DNS (Domain Name System)
- FTP (File Transfer Protocol)

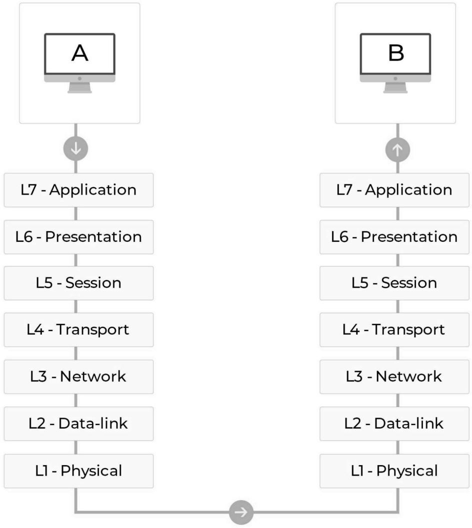
# Communication protocols in web technology

- The most important protocols for data transmission across the Internet are:
  - **TCP (Transmission Control Protocol) and**
  - **IP (Internet Protocol)**.
- Using these jointly (TCP/IP), we can link devices that access the network.
- Some other communication protocols associated with the Internet are POP, SMTP and HTTP.

# Fundamental Concepts in Networking

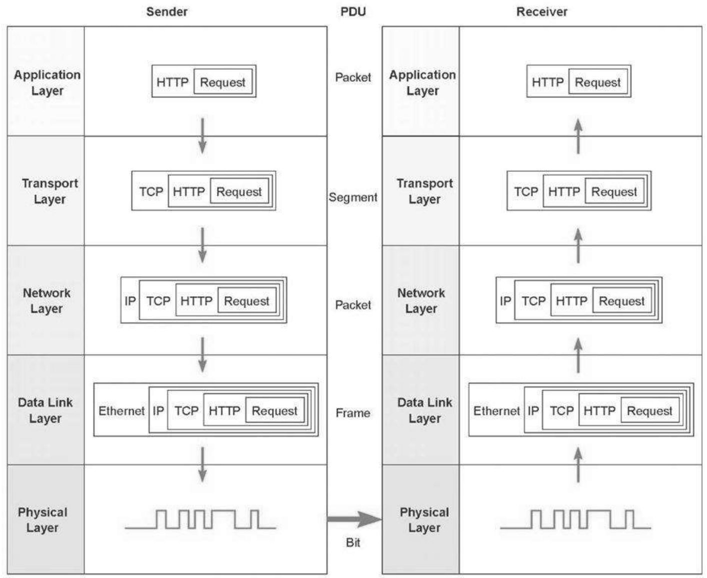


# Network Protocol




## Message transmission using layers.

PDU = Protocol Data Unit
IP = Internet Protocol;
HTTP/Hypertext Transfer Protocol;
TCP = Transmission Control Protocol




## Network models

OSI = Open Systems Interconnection Reference

| OSI Model | Internet Model | Groups of Layers | Examples |
|---|---|---|---|
| 7. Application Layer | 5. Application Layer | Application Layer | Internet Explorer and Web pages |
| 6. Presentation Layer | | | |
| 5. Session Layer | | | |
| 4. Transport Layer | 4. Transport Layer | Internetwork Layer | TCP/IP Software |
| 3. Network Layer | 3. Network Layer | | |
| 2. Data Link Layer | 2. Data Link Layer | Hardware Layer | Ethernet port, Ethernet cables, and Ethernet software drivers |
| 1. Physical Layer | 1. Physical Layer | | |

## References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# The End